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The effect of social interaction in virtual meeting settings on work engagement: a socio-technical systems study

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Abstract
The COVID-19 pandemic and the resulting lockdowns drove new types of interactions between employees. Before the pandemic, social interaction could be performed both in person and online, but measures against COVID-19 required a reduction in personal contact. As a result, employees were forced to interact via virtual meetings, which influenced their work engagement. Thus, it is important to investigate how social interaction influences work engagement in a virtual meeting context and how virtual meetings should be conducted to promote work engagement. This research adopts socio-technical systems theory in the context of examining the influence of social interaction during virtual meetings on work engagement. We surveyed 195 office employees, who were asked to assess their work engagement according to the perceived technical aspects and social interaction in virtual meetings. Our results show that social and technical aspects indeed influence work engagement. The results enhance our understanding of how to design virtual meetings that support employees in their work and improve social communication with colleagues, which is important for future business communication.

Keywords Work engagement · Social interaction · Virtual meetings · Social exchange · Social communication · Socio-technical systems theory

1 Introduction

Virtual meetings emerged as a sudden substitute for almost all in-person work meetings during the COVID-19 pandemic, when measures were implemented to reduce in-person contact (e.g., in Germany [23],. Individuals who were forced to work from home became increasingly reliant on virtual meetings [53]. While some individuals were already experienced with virtual meetings, many employees had never engaged in this meeting type previously and had to quickly familiarize themselves with this format [3, 66]. Although many studies have explored the impact of working from home on work engagement during the pandemic, they have not specifically examined the role of social interaction in virtual meeting contexts from a socio-technical perspective [22, 51]. Addressing this research gap is important as the interplay between social and technical systems influences work engagement and performance [7, 8, 51]. Further, social interaction in virtual meetings is of particular significance as the technical aspects of these meetings, wherein participants are distributed all over the world, are persistent and inevitable [64]. Specifically, insufficient social interaction (e.g., due to interruptions; Zaw & Thaw [65], can lead to misinterpretations, which can hinder further social interaction [41]. Thus, this study utilizes socio-technical systems (STS) theory [7, 8] to answer the following research question: *What is the effect of social interaction in virtual meeting settings on work engagement?*

To answer the research question, we use data from a questionnaire conducted with 195 employees and analyze

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our proposed research model. We investigate the influence of social interaction on information, decision, and brainstorming meetings, both for individuals who participated in virtual meetings before the COVID-19 pandemic and for those who did not. Our results contribute to the understanding of social interactions in virtual meetings and how to conduct virtual meetings to promote work engagement. Thus, our results can be used to improve future business communication.

The remainder of this paper is organized as follows. Section 2 describes the theoretical background and literature review of our study, followed by the theoretical development in Sect. 3 and hypotheses development in Sect. 4. In Sect. 5, we present our materials and methods. Section 6 outlines the results. In Sect. 7, we discuss the research findings and provide theoretical contributions, practical applications, as well as limitations and future research directions. Finally, in Sect. 8, we conclude our study.

2 Theoretical background and literature review

2.1 Virtual meetings

Virtual meetings are conducted by individuals who are geographically separated from each other and must therefore communicate, synchronously and asynchronously, using technology [16, 42]. Such technologies facilitate voice and video calls or even virtual meeting spaces with avatars [21]. Companies use virtual meetings as formal arrangements with specific goals, comparable to face-to-face meetings [53]. To conduct a virtual meeting efficiently, appropriate selection of the meeting type is important [1, 53]. Of the several meeting types, we focus on work-related virtual meetings, specifically information, decision, and brainstorming meetings, respectively [1]. (Virtual) information meetings serve to present or sell an idea, product, service, or project. (Virtual) decision meetings also have a single purpose, focusing on a problem that needs to be solved or a decision that needs to be made. (Virtual) brainstorming meetings are dedicated to generating new ideas or concepts [1].

2.2 Work engagement

Work engagement is defined as a positive and fulfilling attitude toward work-related conditions [55]. Thereby, engagement is associated with the terms vigor, dedication, and absorption. Vigor entails a high level of energy combined with endurance; dedication refers to pride, challenges, and a sense of importance; and absorption reflects the ability to concentrate while working on one's tasks [55]. An employee is engaged if certain surrounding work conditions trigger extrinsic or intrinsic motivation, which is subsequently strengthened if tasks are completed successfully. Through such activation, a feeling of enthusiasm can be evoked in an employee accompanied by identification with the work, leading to a feeling of dedication [55].

2.3 Social interaction in virtual meeting settings

To understand social interaction and work engagement in a virtual meeting context, social communication during virtual meetings and the general social exchanges between employees have to be considered [56, 60]. First, concerning social communication during virtual meetings, Shaik and Makhecha [56] investigated drivers of employee engagement in global virtual teams. They determined that social communication during virtual meetings improves work engagement and performance. Further, they stated that interpersonal relationships can be created and nurtured using informal social communication during such meetings, but that this is dependent on the social skills of the participants, who have to be aware of the cultural backgrounds of other participants and be empathic to their needs. When workers are physically isolated, such as during a pandemic, social communication in virtual meetings is essential as ad hoc conversations that would emerge by physically bumping into colleagues by coincidence do not take place. Kelley and Kelloway [38] stated that the rules of conversation during virtual meetings and appropriate social communication mandate that participants see each other onscreen; otherwise, virtual meetings are less efficient than face-to-face meetings, and misinterpretations could emerge that negatively impact work engagement. Chanana and Sangeeta [13] supported this proposition by including it in a list of appropriate employee engagement practices derived from experiences during the COVID-19 lockdown. Another important rule of conversation is that overlapping speech should be avoided, as it can also lead to misinterpretations that lower work engagement [65]. Karl et al. [36] observed that the inclusion of survey and chat tools during virtual meetings has positive effects on employee engagement. They showed that frustrations arising during virtual meetings are caused by the same aspects as those arising during face-to-face meetings. Nonetheless, individuals perceive that even social chatting using digital platforms is subject-driven, meaning that spontaneous social communication is limited [64]. This proposition was supported by Chan [12], who determined that subject-driven conversation and discipline are important aspects of conducting virtual meetings during pandemics. Thus, companies should enact measures to encourage social chatting between employees and to maintain social exchanges in general during pandemics; e.g., by organizing

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virtual social events [64]. Such measures are appropriate to reduce feelings of social isolation that persist during pandemics [4, 22]. Further, Chen and Wei [14] stated that social communication using social media positively influences employee performance.

Second, alongside social communication during virtual meetings, social exchange in general is necessary to influence work engagement and performance [19, 64]. Driskell et al. [19] investigated the effects of technological mediation on team performance and determined that social interaction is mandatory for successful teamwork in virtual collaborations. Further, Goel et al. [24] showed that attachment—i.e., how actively an individual pursues or avoids social interaction in a team context—positively influences work engagement. Concerning work intensity, Nagata et al. [47] investigated the effect of different intensity levels on work engagement. They determined that low and medium work intensities—meaning one to three days of work per week—raised work engagement during the COVID-19 pandemic. To compare the between-person and within-person impacts of telework in teams, Delanoeije and Verbruggen [18] conducted a study in a Belgian company, where one team worked from home two days a week and the other did not work from home at all. They determined that the team working from home experienced lower stress, a better work–life balance, and higher job engagement and performance on teleworking days compared to non-teleworking days. In their study addressing pre-COVID-19 teleworkers, Park and Jae [49] determined that work similarity, telework frequency, the accessibility of technologies, and the quality of performance management positively affected the organizational impact of telework. Further, telework frequency, the quality of performance management, and the quality of supervision as a social interaction were positively related to job performance. Chen and Wei [14] showed that social media use improves social exchange relationships and thus the performance of employees. Chun et al. [15] conducted a multilevel study of transformational leadership, social exchange relationships, and performance in teams and showed that social exchange in general is important for team performance. Torrente et al. [62] investigated how teamwork engagement mediates between social resources and performance in teams. They determined that communication during virtual meetings is positively related to work engagement, while both communication and work engagement foster team performance [62]. These conclusions were also supported by Tekleab and Chiaburu [59], who determined that a good working atmosphere via general social exchange is necessary to improve work engagement. Additionally, in a review of the literature concerning enhancing the effectiveness of virtual teams, Berry [6] pointed out that social interaction is important. This proposition was confirmed by Newman et al. [48], who analyzed responses from different virtual teams and determined that team members who perceived that their leaders communicate effectively also perceived that their team's performance improved. Rayton and Yalabik [43, 52] determined that social exchange in general increases cohesion between employees in virtual teams and improves team performance. Further, appropriate formal and informal social exchange in general fosters work engagement among virtual team members [56]. A study by ten Bummelhuis et al. [60] examined work engagement, fatigue, and communication quality in virtual teams. The results showed that working in virtual teams has a positive effect on daily engagement and a negative effect on daily exhaustion. Shaik and Makhecha [56] revealed that working in virtual teams leads to higher work engagement when members have a high degree of maturity and appropriate formal and informal communication cultures—i.e., high levels of social exchange in general. In addition, Bennett et al. [4] indicated that employees often show signs of fatigue during long virtual meetings, which lowers their work engagement. Waizenegger et al. [64] argued that virtual meetings are often used to control employees while they are working from home, which also lowers their work engagement. Whereas studies have determined that employees' work engagement and work–life balance are positively influenced by participating in virtual meetings, they have also shown that loneliness and inefficient communication increase due to meetings taking place in this form [22, 51].

2.4 Technical interaction in virtual meeting settings

It is essential to consider the perceptions of technical aspects during virtual meetings, as they also influence work engagement and performance [5]. Kayworth and Leidner [37], Martins et al. [44], and Morris et al. [45] all pointed out that in order for leadership to build trust relationships among virtual team members, seeing each other on screen is mandatory. Hence, all of these studies proposed requiring virtual meeting participants to use their webcams during virtual meetings to improve leadership measures, trust relationships, and team performance. Further, Zhiqiang and Abbas [66] examined the psychological challenges of virtual meetings in regard to employee resilience. If employees who otherwise work closely together cannot see each other, work engagement is negatively impacted, which endangers employees' mental health [66]. Carnevale and Hatak [9] supported this proposition by investigating employee adjustments and well-being in the era of COVID-19, as did Müller et al. [46] in their investigation of factors that cause information overload in virtual team settings. Additionally, Thompson et al. [61] gathered features for a virtual meeting prototype design and determined that location independence

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is mandatory for virtual meeting success. Hence, participants should be allowed to conduct the meeting in their location of choice.

3 Theoretical development

3.1 Socio-technical systems theory

To analyze the impact of social interaction on work engagement in a virtual meeting context, we adopted the STS theory developed by Trist and Bamforth [63]. According to STS theory, individuals use technologies to produce products or services that in turn influence those individuals and the interactions between them [50]. A successful system consists of a combination of social and technical components that influence each other and work symbiotically [11, 58]. The social component focuses on interactions between individuals, along with their attitudes and skills. The technical component refers to the processes and technologies used to produce the products and services [7]. A system should focus on both aspects to achieve effective performance [17, 28, 58]. If STS are applied appropriately, failures and problems in the management of information systems are reduced and, as a result, the management of information systems is improved [7]. In the past, STS theory has proven to be successful in aligning social and technological factors in companies to improve collaboration and engagement and increase companies' success [20]. Hence, we use STS theory to investigate the role of social interaction in virtual meeting settings to improve work engagement and, ultimately, to improve work performance.

3.2 The STS perspective in virtual meeting settings

Various past studies have investigated the effects of social interaction on work engagement and performance for different types of virtual collaboration, such as virtual meeting settings, virtual teams, and remote and telework settings. Whereas most studies prior to COVID-19 focused on virtual collaboration in virtual team settings, studies that have determined factors for collaboration in virtual meeting settings in general have since gained prominence. Due to the COVID-19 pandemic and resulting curfews, virtual meeting settings have become an effective way to collaborate with individuals outside of a team, such as customers or suppliers [2, 21, 39]. Investigating such scenarios is important, as prior to COVID-19, many collaboration processes with customers or suppliers were held in-person [64], while collaboration in virtual teams was already well-investigated (e.g., [5, 6, 19]),. Whereas certain disagreements, such as connection difficulties in online meetings, are better tolerated when collaborating within a team, this is not the case when collaborating between teams, as meetings between the team and customers or suppliers, for example, should run as smoothly as possible [64]. Hence, to ensure that collaboration runs as smoothly as possible, social and technical interaction and especially their interaction in virtual meetings should be considered [17, 28, 43, 58]. It is therefore necessary to determine which factors of social and technical interaction support successful virtual meeting settings—i.e., improve work engagement [7]. Nonetheless, there has been very little research on the interplay between the social and technological systems of virtual meeting settings. As several studies point as that sufficient interplay between social and technological systems of virtual meetings in teams and between teams is crucial for companies' success [14, 51, 64], this interplay is important to investigate. In conducting such research, STS can be effective as a guiding theory because it recognizes that social interaction is mandatory to structure people, technology, and tasks [64]. Hence, our investigation seeks to explain how people, technology, and tasks should be aligned in virtual meeting settings in order to improve the work engagement of employees.

4 Hypotheses development

In virtual spaces, social interaction is composed of social communication [5, 6, 14, 48, 62] and social exchange [15, 52, 59, 62]. Regarding social exchange in general, maintaining a good working atmosphere that supports social exchange has a positive influence on work engagement [59, 62]. If employees feel valued in their company due to receiving support for social exchange, their work engagement is higher [52]. The quality of information being exchanged impacts a virtual meeting's success, but social communication is also important and enhances meeting performance [14, 15]. STS supports this proposition as it states that social systems are not only concerned with the attributes of people but also the relationships among them, the reward systems, and the authority structures [7]. Thus, our first hypothesis is:

H1 *Increased social exchange in general among employees during daily work equates to increased social communication among these employees during virtual meetings.*

Social communication during virtual meetings increases feelings of cohesion between employees and ultimately leads to success [16]. This is supported by STS theory, as it is the secondary users, not the designers of the management information system, who decide whether the system is successful [7]. Hence, our second hypothesis is:

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H2 *Increased social communication among employees during virtual meetings equates to increased work engagement among these employees.*

Social exchange between employees in general positively influences their work engagement [52]. Subsequently, if social exchange between employees in general is interrupted or blocked, their work engagement is reduced [36, 52, 55]. STS theory supports this proposition, as the interplay between people, structure, and technology determines how successful tasks are; i.e., how work engagement during all tasks is improved [7]. Thus, our third hypothesis is:

H3 *Increased social exchange in general among employees during daily work equates to increased work engagement among these employees.*

Eye contact is essential for effective social communication during virtual meetings. In the absence of face-to-face communication nonverbal cues are reduced or lost, which can reduce company cohesiveness and give rise to misinterpretations [38]. While video tools such as Zoom allow employees to see each other, mutual eye contact during virtual meetings is still reduced, resulting in poorer communication in virtual meetings versus face-to-face meetings [13]. Further, it is important that employees can interact in virtual meetings without long delays, as social communication is otherwise limited [65]. Both the lack of eye contact and long delays could negatively influence social communication. STS supports these propositions, as technological aspects have to be sufficient in order to benefit social interaction [7]. Hence, our fourth hypothesis is:

H4 *The more positively technical aspects in virtual meetings are perceived by employees, the greater the social communication among these employees during virtual meetings.*

In virtual meetings, satisfaction with technical aspects has a positive impact on the job satisfaction of the virtual meeting members [37, 38, 44, 45, 61]. Researchers have identified two important technical aspects of virtual meetings. First, just as in face-to-face meetings, a critical factor in virtual meetings is visual, face-to-face communication. Face-to-face communication enhances relationships and can be facilitated in virtual meetings via tools such as webcams [37, 38, 44, 45]. Second, location independence is an important technical aspect in virtual meetings. Members do not have to spend time or money to travel to the meeting location, which is particularly important for employees who have meetings with colleagues in locations other than their common workplace [61]. Thus, independent locations have been especially relevant in the context of the COVID-19 pandemic, as the number of people working from home has increased [39]. Again, STS theory supports these propositions as the ultimate function of management information systems is to support collaboration; i.e., to improve work engagement in virtual meeting settings [7]. Thus, our fifth hypothesis is:

H5 *The more positively technical aspects in virtual meetings are perceived by employees, the greater the work engagement among these employees.*

Figure 1 summarizes the hypotheses in the research model.

5 Materials and methods

5.1 Questionnaire

We designed an initial questionnaire to test our research model that is provided as additional material for this study. Due to statistical tests that are described in Sect. 5.3, we had to reduce the number of items in our initial questionnaire, so that our final questionnaire is shown in Appendix A-1. We utilized a 5-point Likert scale format for each item, ranging from 1 ("Do not agree at all") to 5 ("Completely agree"). The items for the formative variables "social communication during virtual meetings" (2 items) and "perceived technical aspects in virtual meetings" (2 items) were derived from important characteristics of virtual meetings concerning STS [27] and drivers of employee engagement in global

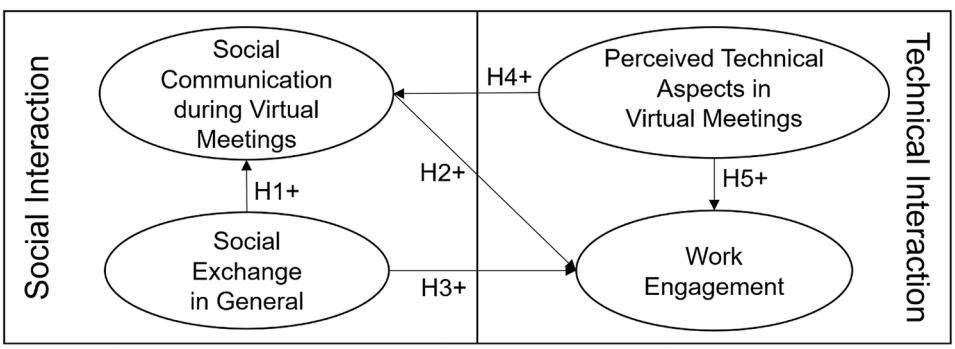


Fig. 1 Research model

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virtual teams [56]. Both studies point out that communication in virtual meetings should be as like as face-to-face meetings, with the advantage of participating location-independently. The reflective variable "work engagement" (9 items) was measured using the questionnaire from Schaufeli et al. [54], while "social exchange in general" (4 items) was measured using the scale from Chen and Wei [14].

5.2 Sample

To gather our participants, we used the unsupervised crowdworking platform Clickworker (similar to Amazon MTurk). Following the recommendations on using such platforms from Goodman et al. [25], we used attention checks and kept the questionnaire short. Further, we included test questions at the beginning and the end of the questionnaire to verify that participants' self-reported statuses were correct. We focused on employees without managerial responsibilities because they do not typically set up required meetings; this task is usually carried out by managers who are aware of the goals, resources, and processes of the company [3]. Hence, we obtained data on meeting participants rather than organizing hosts who are able to control how the meetings are conducted [27]. We surveyed employees who lived in Germany in September 2020. This point in time was considered to allow us to capture a situation in which employees had had sufficient time to experience virtual work environments while still perceiving them as temporary. We gathered responses from a total of 195 participants, of whom 36.92% ($n=72$) were female and 63.08% ($n=123$) were male. The average age was 37.52 years (SD: 10.09). Our participants worked in departments including customer service (15.90%, $n=31$), IT (10.77%, $n=21$), finance and accounting (8.72%, $n=17$), production (8.21%, $n=16$), sales (7.69%, $n=15$), marketing (6%, $n=13$), project management (5.64%, $n=11$), audit/quality control (4.10%, $n=8$), human resources (3.59%, $n=7$), logistics (3.59%, $n=7$), product development (3.59%, $n=7$), process engineering (3.08%, $n=6$), procurement (2.56%, $n=5$), product management (2.05%, $n=4$), company development (0.051%, $n=1$), and others (13%, $n=26$). A total of 47.18% ($n=92$) of the respondents had participated in virtual meetings prior to the COVID-19 pandemic, while 52.82% ($n=103$) attended virtual meetings for the first time during the COVID-19 pandemic. Concerning the amount of time working from home in relation to total working time, 34 participants did not work from home at all, 44 participants worked from home 20% of the time, 31 worked from home 40% of the time, 23 worked from home 60% of the time, 28 worked from home 80% of the time, and 35 worked from home 100% of the time (mean: 67.38%; variance: 61.89%; SD: 35.18%).

5.3 Statistical tests

We used partial least squares structural equation modeling (PLS-SEM) to test our research model. We applied SmartPLS 3.3.9 to estimate our weightings with a path method and determine the significance of the path coefficients using bootstrapping procedures with 5,000 samples [29]. To verify our model, we conducted several statistical tests. First, we followed the requirements of Hair et al. [29] and Hulland [35] to test the reflective variables "social exchange in general" and "work engagement." The values for indicator reliability were above 0.7 (see Appendix A-2), as were the values for composite reliability 0.7 (see AppendixA-4). The average variance was higher than 0.5 (see AppendixA-4; Hair et al., [29], and the values for the discriminant validity using the heterotrait–monotrait ratio were below 0.9 (see AppendixA-5; Henseler, Ringle, et al., [32]).

Second, we tested the formative variables "social communication during virtual meetings" and "perceived technical aspects in virtual meetings," as follows. Multicollinearity was deemed to have been avoided as the variance inflation factors were below 5 (see AppendixA-3; Hair et al., [29]. Further, the relative and absolute importance of indicators were examined, with loadings and weights that were all significant (see AppendixA-3). We then had to determine whether the bivariate correlations were higher between an indicator and the variable than between indicators [10] to check heterogeneity. We concluded that no suppressors and no collinear indicators were present.

We validated the quality of our structural model using several tests. First, the standardized root mean square residual was used to determine the approximate fit of our composite model, as well as the common factor model (Henseler, Dijkstra, et al., [32]. We obtained values of 0.071 for both the composite and the common factor model. Further, we conducted a blindfolding procedure according to Kortmann [40], with an omission distance of 7, to assess the prediction relevance of the model [34]. As we identified positive Stone–Geisser Q^2 values for our model (see AppendixA-6), per Henseler et al. [34] we determined that an appropriate predictive power had been obtained. As the last of our statistical tests, we performed PLS predictions according to Shmueli et al. [57]. Our PLS-SEM Q^2 predicted values were solely positive, and the prediction errors were not highly symmetrically distributed. Thus, we checked whether our PLS-SEM mean absolute errors (MAEs) were higher than those in the linear regression model (LM). As the minority of our PLS-MAEs were higher than the LM ones (see AppendixA-8), we determined a low predictive power of our model, per Shmueli et al. [57].

Regarding common method bias, we followed the recommendations of Guide et al. (2015), who have downplayed

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Table 1 Descriptive results

	M	SD	(1)	(2)	(3)	(4)
(1) Social Exchange in General	3.41	0.62	-	0.260**	0.096	0.509**
(2) Social Communication during Virtual Meetings	3.47	0.64		-	0.488**	0.299**
(3) Perceived Technical Aspects in Virtual Meetings	3.94	0.82			-	0.276**
(4) Work Engagement	3.61	0.69				-

Notes: M = mean; SD = standard deviation, ** $p < 0.01$

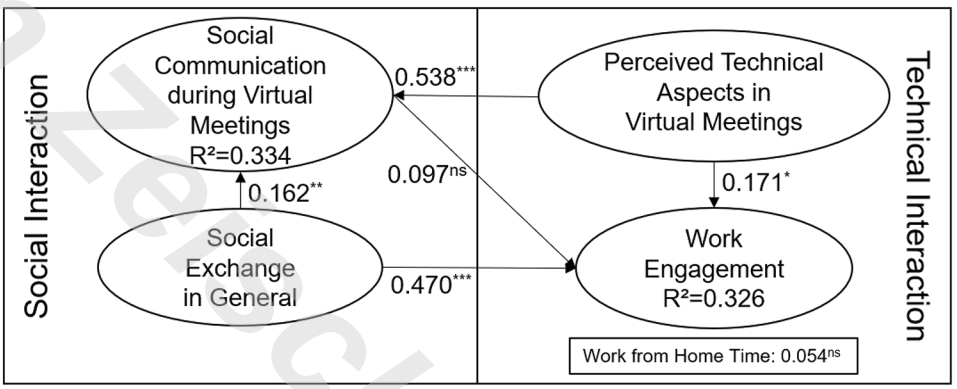


Fig. 2 Results of the research model. Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; one-tailed tests

the usefulness of conducting common approaches such as Harman's test. Since such tests do not allow for the conclusion of an absence of common method bias, we took careful measures with the design of the questionnaire as described above.

6 Results

6.1 Descriptives

The descriptives and correlations among variables are shown in Table 1.

6.2 Results regarding the research model

The results in relation to each hypothesis are depicted in Fig. 2, including "work from home time" as the lone control variable. We decided to not use age and gender as control variables, as our participants were rather young and predominantly male.

As can be seen in Fig. 2, our results empirically support Hypothesis 1, meaning that social exchange in general influences social communication during virtual meetings (0.162, $p < 0.01$, $f^2 = 0.039$). In contrast, for Hypothesis 2 our results reveal that social communication during virtual meetings does not influence work engagement (0.097; $p > 0.05$; $f^2 = 0.009$). For Hypothesis 3, our results support the notion that social exchange in general positively influences work engagement (0.470, $p < 0.001$; $f^2 = 0.312$). Further, our results reveal that for Hypothesis 4, perceived technical aspects in virtual meetings positively influence social communication during virtual meetings (0.538, $p < 0.001$; $f^2 = 0.430$). Regarding Hypothesis 5, our results confirm that perceived technical aspects in virtual meetings positively influence work engagement (0.171, $p < 0.05$; $f^2 = 0.030$). The variance of work engagement is explained with 32.6% ($R^2 = 0.326$) in our model, which is considered "moderate" per Hair et al. [30].

Additionally, we determined that our control variable, "work from home time," does not influence work engagement. We also conducted post hoc analyses by separating our questionnaire data concerning the different meeting types (information, decision, and brainstorming meetings, respectively), but we were not able to determine further significances (see AppendixA-7). We then examined the questionnaire data by separating individuals into two groups: those who had already attended virtual meetings before the COVID-19 pandemic and those who had participated in virtual meetings for the first time during the pandemic. Again, we were unable to determine further significances (see AppendixA-7).

7 Discussion

This study adopted the socio-technical systems theory [63] to investigate the role of social interaction on work engagement in a virtual meeting context. In virtual meetings, social communication [31, 42, 62, 65] and the general social exchange that takes place between employees [21] are important [13]. Further, the technical aspects of virtual meetings have to be considered in order to support social interaction [2, 4, 12, 36, 37, 44, 45, 61].

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7.1 Research findings

Regarding the impact of social interaction on virtual meetings and technical aspects on work engagement, our results reveal that a third of work engagement's variance is explained. This positive impact of social interaction was also found by Galanti et al. [22], who determined that social isolation among employees during lockdowns led to lower work engagement. Concerning our hypotheses, we found support in our data for Hypotheses 1, 3, 4, and 5, but not for Hypothesis 2.

As for Hypothesis 1, we confirm that social exchange in general positively influences social communication during virtual meetings. Thus, leaders and companies should support social exchange between colleagues [6, 16, 46, 48, 52] to improve social communication during virtual meetings. This is especially relevant during lockdowns, as found by Tekleab and Chiaburu [59] and Chanana and Sangeeta [13] for employees and by Gabbiadini et al. [21] for individuals in general.

Concerning Hypothesis 2, our results do not support the idea that social communication during virtual meetings improves work engagement. Hence, our results indicate that considering social customs such as bilateral chats, rules of conversation or purposeful statements during virtual meetings does not affect the level of work engagement of each employee. Thus, our results contradict those of Torrente et al. [62], who determined that work engagement is positively influenced by social communication and ultimately leads to better team performance; nor do they align with Zaw and Thaw [65] determination that overlapping speech hinders social communication during virtual meetings and lowers work engagement. Our results also contradict the proposition of Karl et al. [36] that the use of chat tools as a form of social communication during virtual meetings positively influence employees' engagement. Additionally, our results contradict the claims of Shaik and Makhecha [56] in that social communication during virtual meetings does not strengthen relationships and that informal conversation is not improved. Thus, the absence of empathy toward other participants and a lack of knowledge about the cultural backgrounds of other participants might be reasons for the lack of support of social communication during virtual meetings toward work engagement. On the other hand, our results support the propositions of Waizenegger et al. [64] in that social communication during virtual meetings is rather subject-driven and limited. Thus, we conclude that due to many meetings taking place during COVID-19 there has been a shift in the needs of social communication during virtual meetings in contrast to studies conducted prior COVID-19, such as of Torrente et al. [62] and Shaik and Makhecha [56]. Further, we propose that due to the large amount of such meetings during COVID-19, social communication takes place next to the meetings, so there is no urgent need for chatting during the meeting. Additionally, as virtual meetings substituted almost any form of conversation during COVID-19, participants are not hindered by overlapping speeches, as they are a natural part of frequent conversations, such as in face-to-face meetings. Hence, we do support the statement of Waizenegger et al. [64] that companies should conduct virtual social events to encourage relationships and informal conversation among employees.

Regarding Hypothesis 3, our results reveal that social exchange in general positively influences work engagement. Thus, we support the proposition of Chun et al. [15] that social exchange positively influences team performance. Further, our findings align with those of Chen and Wei [14], who highlighted the importance of social media for general social exchange to counteract social isolation.

As for Hypothesis 4, our results show that perceived technical aspects in virtual meetings positively influence social communication during those meetings. Hence, to enable social communication to happen in the first place, it is necessary to see colleagues onscreen in lieu of face-to-face meetings. Thus, we support the propositions of Baker and Murphy [2], Chan [12], Karl et al. [36], and Kelley and Kelloway [38] regarding the importance of colleagues seeing each other to enable sufficient social communication. Further, technology should facilitate the location-independent participation of employees in virtual meetings to enable their social communication. Therefore, our findings support the insights of Thompson et al. [61], who stated that location independence is important for virtual meeting systems.

Concerning Hypothesis 5, our results confirm that perceived technical aspects in virtual meetings positively influence work engagement. Thus, our findings support the propositions of ten Bummelhuis et al. [60], who determined that technological innovations that make it possible to work from any location positively influence work engagement.

Hence, we conclude that social interaction influences work engagement in a virtual meeting context, while work engagement is influenced by social exchange only. As social communication does not influence work engagement, and although employees bilaterally chat during virtual meetings, we determined that social exchange in general is the main factor impacting social interaction. Thus, we confirm the proposition of Carnevale and Hatak [9] that social interaction and social exchange in general between employees are important topics to consider for human resource management. Concerning STS, our results reveal that both social and technical aspects have to be considered to enhance meeting performance—i.e., social interaction between employees is necessary to conduct better meetings [17, 28, 43, 58]. Hence both, social and technical aspects should be

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considered in the design of future virtual meetings to conduct them sufficiently—i.e., to align technology, people, and tasks—as STS theory states [7]. Although we conducted post hoc analyses on our data set concerning different meeting types (information, decision, and brainstorming meetings, respectively), we were unable to determine further significances. Thus, our results support the proposition of Rubinger et al. [53] that social interaction is important during lockdowns and should be facilitated in every type of meeting. We were likewise unable to determine further significances between individuals that had attended virtual meetings before the COVID-19 pandemic and those who had only begun to attend virtual meetings since then. Hence, we support the insights of Gabbiadini et al. [21] that social interaction during lockdowns is relevant for anyone who encounters social isolation.

7.2 Theoretical contributions

Our study contributes to the information systems literature in several ways. First, we introduce STS theory to investigate the interplay of social and technical systems on work engagement in a virtual meeting context, which is important to improve companies' success, according to Chen and Wei [14], Pass and Ridgway [51], and Waizenegger et al. [64]. As the theory itself is rather abstract and requires more adaptation to the context compared to other theories, we contribute conceptually to its application level in our given context. Specifically, we highlight how technical aspects in virtual meetings and social interaction during such meetings, as well as in general, interact in their influence on work engagement.

Second, our empirical results add to the knowledge of how work engagement is influenced by social interaction in a virtual meeting context. Whereas related studies have focused on social interaction in broader terms, for instance by investigating it as part of strategies for virtual collaboration success [5] or determining how it is influenced by social media in times of social isolation [14], we take a different approach. We provide insights into the importance of work engagement, especially in times of social isolation, which is important, as social isolation lead to disruptive changes in work environments [64]. Hence, our results are expected to contribute to improving employees' work engagement in a virtual meeting context, especially in future times of isolation.

Third, we provide social and technological requirements for future virtual meetings, which is important for future virtual meetings and business communication research to be effective, due to social interaction in STS [6, 48, 58]. Hence, our results can be used to improve study designs concerning virtual meetings and business communication.

Fourth, we determine that with respect to social interaction and its effects in a virtual meeting context on work engagement, only social exchange in general is relevant to predict work engagement, with social communication during virtual meetings having no influence. Thus, unlike prior studies, we conclude that with regard to work engagement in virtual meeting contexts, social exchange in general is more important than social communication.

7.3 Practical applications

Concerning practical applications, we recommend that companies support social interactions between employees and consider technical aspects regarding virtual meetings, as these are relevant factors for work engagement among employees. First, our results show that location-independent participation and the desire to see each other are important technical aspects of virtual meetings, and positively influence work engagement. Thus, companies should allow their employees to choose their location for participating in virtual meetings and motivate them to turn on their webcams during such meetings to improve their work engagement.

Second, as we did not find any significance concerning our control variable or in our post hoc analyses, we strengthen our first implication for virtual meetings in general. This is because, no matter how often employees worked from home, what kind of virtual meeting they attended, or whether they had ever participated in virtual meetings before the COVID-19 pandemic, work engagement was not affected in our study. Thus, companies should allow employees to choose their location for any kind of virtual meeting and every employee alike, no matter how often they work from home or how much experience they have with virtual meetings.

Third, as social exchange in general is important for work engagement, we propose that companies analyze their employees' problems and potential. Specifically, companies should motivate their employees to support each other via positive communication. If one employee does something that facilitates the work of other employees, such actions should be celebrated.

7.4 Limitations and future research

As with any research, our analysis is subject to limitations. First, we used the crowdworking platform Clickworker to gather our participants online. While studies have shown that results between voluntary and paid data-gathering are similar (e.g., Goodman et al., [25]), results might differ for voluntary participants as online crowdworkers earning money parallel to their main job might be more used to virtual work in general. Second, as we used social interaction

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and technical aspects during virtual meetings as predictors for work engagement, our propositions are limited to socio-technical contexts only. Thus, there could be other factors that influence work engagement, and these should be considered in further studies. Third, we gathered our study data at one point in time from informants who may have been subject to bias. While we applied measures to prevent this, other studies should investigate possible long-term effects on work engagement.

Concerning future research, social interaction should be addressed for different kinds of virtual collaboration. As such, developments in virtual meetings require the investigation of new possibilities for social interaction, such as meeting with others in a metaverse. In addition, further research into the effects of social interaction on work engagement in a virtual meeting context in times without forced social isolation would be interesting.

8 Conclusion

In this study, we adopted STS theory [63] to investigate the relationship between social interaction and work engagement in a virtual meeting context. As our results reveal that social interaction between employees and perceived technical aspects in virtual meetings positively influence work engagement, we conclude that STS theory is appropriate to investigate the phenomenon. Hence, our study helps companies to improve their business communication in the future.

Appendix

A-1 Questionnaire

Perceived Technical Aspects in Virtual Meetings (PTAVM)
-1- "It is important to me that I am able to see the others on screen in my virtual meetings."
-2- "It is important to me that I can participate in virtual meetings location-independently."
Social Communication during Virtual Meetings (SCVM)
-1- "In virtual meetings, rules of conversation are observed, so that I can participate in conversations without long delays."
-2- "Concerning rules of conversation, I can engage in virtual meetings like I can do in face-to-face meetings."
Social Exchange in General (SEG)
-1- "My colleagues inform me if I do something that facilitates their work."
-2- "I inform my colleagues if they did something that facilitates my work."
-3- "My colleagues know my potential very well."
-4- "My colleagues understand my problems very well."
Work Engagement (WE)
-1- "During my work, I feel myself full of energy."
-2- "In my job, I feel myself strong."
-3- "I am keen concerning my job."
-4- "My job inspires me."
-5- "After getting out of bed in the morning, I want to go to work."
-6- "I am happy if I work intensively."
-7- "I am proud of my work."
-8- "I dig myself into my work."
-9- "I let myself thrill from my work."

A-2 Loadings of reflective variables

Construct	Items	Loadings
SEG	SEG −1-	0.729
	SEG −2-	0.702
	SEG −3-	0.809
	SEG −4-	0.789
WE	WE -1-	0.758
	WE -2-	0.784
	WE -3-	0.818
	WE -4-	0.768
	WE -5-	0.741
	WE -6-	0.691
	WE -7-	0.783
	WE -8-	0.802
	WE -9-	0.833

A-3 Loadings, Weights and VIF values of formative variables

Construct	Items	Loadings	Weights	VIF
PTAVM	PTAVM −1-	0.735	0.596	1.042
	PTAVM −2-	0.812	0.692	1.042
SCVM	SCVM −1-	0.919	0.825	1.057
	SCVM −2-	0.597	0.405	1.057

A-4 Discriminant validity, Composite Reliability and Average Variance Extracted (AVE) values of reflective variables

	SEG	WE	Composite Reliability	AVE
SEG	0.758		0.844	0.575
WE	0.514	0.776	0.932	0.603

A-5 HTMT values.

	1	2	3	
1				*1*
2	0.105			*2*
3	0.130	0.613		*3*

1 = homeoffice time; 2 = social communication generally; 3 = working engagement

A-6 Stone-Geisser-values

	Q^2 (=1-SSE/SSO)
SCVM	0.179

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	Q^2 (=1-SSE/SSO)
WE	0.186

A-7 Post-Hoc Analyses.

	(1)	(2)	(3)	(4)	(5)
working from home time ◊ WE	0.076^{ns}	0.100^{ns}	0.017^{ns}	0.019^{ns}	0.110^{ns}
SEG ◊ SCVM	0.310^{*}	0.113^{ns}	-0.052^{ns}	0.206^{*}	0.135^{*}
SEG ◊ WE	0.445^{***}	0.604^{***}	0.444^{***}	0.469^{***}	0.475^{***}
SCVM ◊ WE	0.060^{ns}	0.232^{*}	-0.136^{ns}	0.095^{ns}	0.085^{ns}
PTAVM ◊ SCVM	0.403^{**}	0.585^{***}	0.750^{***}	0.544^{***}	0.529^{***}
PTAVM ◊ WE	0.155^{ns}	-0.028^{ns}	0.444^{**}	0.107^{ns}	0.259^{*}

(1) information-meeting; (2) decision-meeting; (3) brainstorming-meeting; (4) individuals that participated in virtual meetings before COVID-19; (5) individuals that participate in virtual meetings since COVID-19

A-8 PLS Predict

	LM RMSE	LM MAE	PLS RMSE	PLS MAE	PLS Q^2_predict
WE -1-	0.755	0.599	0.740	0.582	0.202
WE -2-	0.750	0.602	0.733	0.585	0.178
WE -3-	0.878	0.709	0.869	0.703	0.141
WE -4-	0.901	0.717	0.896	0.716	0.142
WE -5-	0.928	0.750	0.929	0.762	0.117
WE -6-	0.857	0.667	0.840	0.662	0.131
WE -7-	0.823	0.656	0.809	0.649	0.200
WE -8-	0.700	0.550	0.701	0.557	0.234
WE -9-	0.815	0.653	0.810	0.658	0.177

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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